Screening the *Formica* hybrid zone for intrinsic incompatibilities: structure-corrected two-locus association and the bidirectional-sorting test

(work in progress)

Work-in-progress note. This document records methods, reasoning and interim results for the intrinsic (Dobzhansky–Muller) arm of Module D, beyond the descriptive among-population summary reported separately. Everything here is structure-corrected but pre-simulation: a neutral, recombination-matched null (Module E) is still required to convert the leads below into excess-over-neutral statements. Numbers may change as that null and the analyses mature.

1 What an intrinsic incompatibility looks like in replicate hybrids

The design is twenty independent, non-migrating hybrid populations, each sorting *F. aquilonia*/*F. polyctena* ancestry out of the same admixture. Against that background an intrinsic incompatibility can leave three distinct traces, and separating them is what organises the analysis.

(1) Unidirectional sorting. An incompatibility in which one parental combination is fitter drives the *same* parent's ancestry toward fixation repeatedly across replicates. This is the per-locus parallelism that Module A measures, and it is not a two-locus problem. Crucially it is also *invisible* to any two-locus test: linkage disequilibrium between two loci is bounded by their allele frequencies, and as either locus approaches fixation the disequilibrium it can carry goes to zero. The more completely a locus sorts unidirectionally, the less LD it can express. Two-locus tests therefore cannot see the most strongly unidirectionally-sorted loci at all; those belong to Module A, and Module D is complementary rather than redundant.

(2) Bidirectional sorting (equal-fitness incompatibility). If the two parental combinations are *equally* fit while the recombinants are not, each replicate independently fixes one or the other compatible combination, chosen by drift. Per locus this looks like strong sorting in *random* directions — some populations toward *F. aquilonia*, some toward *F. polyctena* — i.e. bidirectional sorting. Its single-locus parallelism test is null by construction ($n_{\mathrm{aqu}} \approx n_{\mathrm{pol}}$), so Module A cannot flag it. The signal lives at the pair level (Section 4). Note that the among-population component of two-locus LD is *maximised* exactly here: a locus fixed in the same direction everywhere has no among-population variance, whereas one fixed in different directions across replicates has a great deal — so bidirectional loci are the natural home of the among-population DMI signal.

(3) Coupling maintained within populations. A third trace is elevated two-locus LD *within* populations (Ohta's $D_{IS}^2 > D_{ST}^2$): particular allele combinations staying rarer than expected inside demes. This is the classic within-deme epistatic-selection signature, but it is badly confounded by population structure — in particular by the fact that our samples are workers from few colonies (nine of twenty "populations" are a single nest; nestmates are siblings), so cryptic relatedness inflates apparent within-population LD (a Wahlund effect). Any test of this trace must therefore control for relatedness.

The descriptive among-population screen reported separately established that the *raw* among-population LD between unlinked clusters is dominated by a single genome-wide ancestry gradient (drift/structure), with no pair specificity. The two analyses below remove that confound in two complementary ways — a mixed model that conditions on genome-wide relatedness (Section 2), and a restriction to bidirectional loci that are orthogonal to the ancestry axis by construction (Section 4) — and both surface the same methodological obstacle first: cross-chromosome paralogy (Section 3).

2 Structure-corrected two-locus association (EMMAX)

Rationale and model. The cleanest way to ask whether two loci are associated *beyond* what shared ancestry and relatedness predict is a linear mixed model. We treat one LD cluster's consensus dosage as a quantitative trait and test its association against every other cluster,

$$y_i = \mu + \beta\, x_j + u + \varepsilon, \qquad u \sim N(0, \sigma_g^2 K), \quad \varepsilon \sim N(0, \sigma_e^2 I), \tag{1}$$

where y_i is the focal cluster's dosage across the 164 individuals, x_j a candidate partner, and K a genomic-relationship matrix (EMMAX; Li, Kemppainen, Rastas & Merilä, a method built for LD-clustered genome-wide data). This is a GWAS in which the "phenotype" is one locus and the "markers" are all the others. The variance components are estimated once under the null and reused for every partner, so a whole Manhattan is cheap.

Why one random effect handles both confounds. The leading eigenvectors of K are the ancestry/structure principal components, so conditioning on K removes the genome-wide ancestry gradient — the confound that swamped the descriptive screen — at the individual level. And because K also encodes the sibling relatedness among nestmates, the Wahlund inflation of within-population LD (trace 3 above) is regressed out in the *same* term. A residual peak between two loci is therefore pair-specific association with both the ancestry gradient and the colony structure already removed. Continuous eMLG dosages are used directly, with no hard-calling. Because the scan is symmetric we use *double* leave-one-chromosome-out: both the focal and the partner chromosome are dropped from K, so neither tested locus's own region deflates its association. This recovers a little power over single leave-one-out and leaves the test calibrated ($\lambda \approx 0.98$–1.05). One further point is worth stating, because it runs against intuition: K must be built from the ancestry-*informative* (differentiated) markers, not from a neutral, LD-pruned background. The confound K has to absorb here is the ancestry-sorting axis, which is a *selected* structure carried by those markers; a neutral basis (right for a demographic control such as the BayPass Ω) dilutes that eigenvector and under-corrects, inflating λ to ~ 1.2 and leaking the ancestry gradient back as spurious positive associations. Finally, each focal trait is residualised on the top ten genome-wide PCs before the mixed model: K cannot remove a *secondary* structure axis that is concentrated on the very chromosomes double-LOCO drops, and such axes otherwise surface as hubs (Section 6). Conditioning on the PCs strips the dominant, climate-aligned axes — after it the random-focal control gives essentially no significant partner — while calibration holds. A first, targeted run scanned 51 focal loci (bidirectional candidates, extreme among-population-covariance loci, and controls); peaks were called at a Bonferroni genome-wide threshold, and each is signed (coupling vs repulsion) by its structure-corrected GLS coefficient — both cis and trans hits are recovered.

Reading a Manhattan. The self-peak at the focal locus and its local LD block are the built-in positive control that the pipeline works; the signal is any *additional* peak that is unlinked from the focal locus (different chromosome, or same chromosome > 10 cM). The scan is well calibrated: genomic-inflation λ ranges 0.95–1.02 across focal sets and the background is flat (Fig. 1b, a control locus with only its self-peak). Fig. 1a shows a focal locus with a clean self-peak and a spread of unlinked peaks — exactly the structure-corrected pair-specific association we are after.

The first result is a warning. The top associations are correlated *within* populations (median within-population $r \approx 0.99$), robust to dropping the most influential individuals — so they are neither the ancestry axis nor a few-individual Wahlund artifact. But $r \approx 0.99$ between two unlinked loci, sustained inside every population, is not what an incompatibility makes; it is the signature of paralogy. This is treated next, and it is why the Manhattan in Fig. 1a distinguishes clean peaks (orange) from duplicate peaks (grey crosses): the focal locus's strongest partners are duplicates, its moderate partners are genuine candidates.

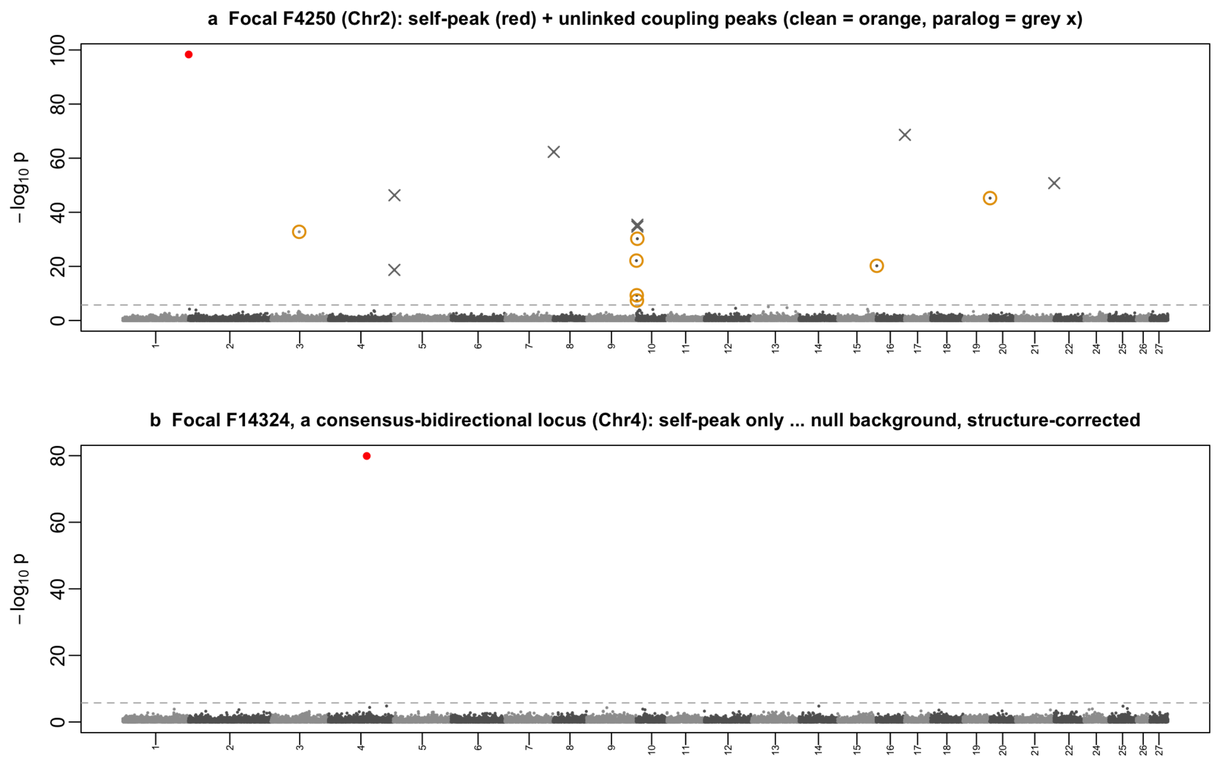

Figure 1: Structure-corrected two-locus association (EMMAX, double leave-one-chromosome-out K from the differentiated eMLGs). **(a)** A focal locus (F4250, Chr2) showing its self-peak (red) and unlinked partner peaks, split by the paralogy filter into genuine candidates (orange) and cross-chromosome duplicates (grey crosses). **(b)** A consensus-bidirectional focal locus (F14324, Chr4): only its self-peak, a flat structure-corrected background ($\lambda \approx 1$). Dashed line: Bonferroni threshold.

3 Cross-chromosome paralogy, and why distance cannot remove it

The problem. Two loci on different chromosomes are at infinite recombinational distance, so any strong LD between them cannot be linkage. It can be a genuine evolutionary correlation — but those are *moderate* — or it can be a technical artifact in which the two "loci" are not distinct at all. When a segmental duplication or a mis-assembly places one genomic region at two locations, reads from the single true region map to both, and the two "clusters" become genotype-identical by construction. Such pairs are the *loudest* inter-chromosomal disequilibrium in any dataset, and they are not biology. In the EMMAX run they reach $r^2 \approx 0.98$ and genotype concordance 1.000, and they recur on particular scaffolds (Chr26). A genetic-distance rule is powerless against them: different chromosomes already are maximal distance. (Distance does its job on the *within*-chromosome side — our conservative complexity-reduction merge at 0.5 cM rather leaves nearby clusters split than over-merges them, and the 10 cM unlinked threshold then excludes those split neighbours from long-range candidacy — but that is a separate problem.)

The discriminant. We flag duplicates by the median *within-population* correlation of the two clusters' dosages. Within-population removes the among-population (ancestry-axis) component, which is not what paralogy is about; what remains is individual-level identity. Genuine long-range coupling in these data sits at $|r| \lesssim 0.6$; a duplication sits at $|r| \approx 1$. A flipped or complementary duplicate (one copy allele-swapped relative to the other) gives $r \approx -1$, which the absolute value still catches. We flag pairs with median within-population $|r| > 0.9$. Two independent signals corroborate the flag: genotype *concordance* approaching 1 (positive duplicates), and per-cluster excess *heterozygosity* — a collapsed duplication makes reads from two paralogs look constitutively heterozygous, inflating H_o (Fig. 2b). The flag is deliberately simple so it can be re-thresholded from the stored values; and it is a screen, not a proof, because an extremely strong genuine within-deme incompatibility could in principle also reach $|r| \approx 1$ — which is why the corroborating signals matter.

It affects both arms, unequally. The filter is shared between the EMMAX and the among-population arms. It flags 32/332 EMMAX hits (10%) but only 55/401,945 among-population survivors (0.01%). The asymmetry is informative: a duplicate *is* structure-corrected individual-level LD, so it surfaces strongly in the mixed-model arm, whereas the among-population statistic — which asks about allele-frequency covariance across populations — is comparatively insensitive to it. In the among-population arm the excess-heterozygosity signal cleanly separates the flagged pairs ($H_o \approx 0.70$ vs 0.37 background); in the EMMAX arm the pre-selected focal set is already high-H_o, so within-population $|r|$ carries the discrimination and H_o is less informative (Fig. 2). Removing the flagged pairs leaves 300 clean EMMAX candidates. The filter is also a free by-product: it identifies duplicated/mis-assembled regions directly from the population data.

4 The bidirectional-sorting test, and cis/trans coupling

Why bidirectional loci, and why at the SNP level. An equal-fitness incompatibility (trace 2) is exactly the case the among-population statistic is built for, and it has a decisive advantage: bidirectional loci do *not* track the genome-wide ancestry gradient — that is what makes them bidirectional — so the confound that dominated the general screen is largely absent, and their co-sorting is a comparatively clean signal. Bidirectional sorting is real at the SNP level (911 SNPs) but is averaged *below* the sorting threshold at the eMLG consensus (no cluster qualifies): averaging member markers pulls the fixation fraction toward the middle. We therefore work at the SNP level, taking each LD cluster's most bidirectional member SNP as its representative (599 reps), which de-redundifies without losing the signal. One chromosome stands out — Chr24 carries 176 of the 911 bidirectional SNPs, but these collapse to only 13 clusters, i.e. a single large LD block (supergene-like).

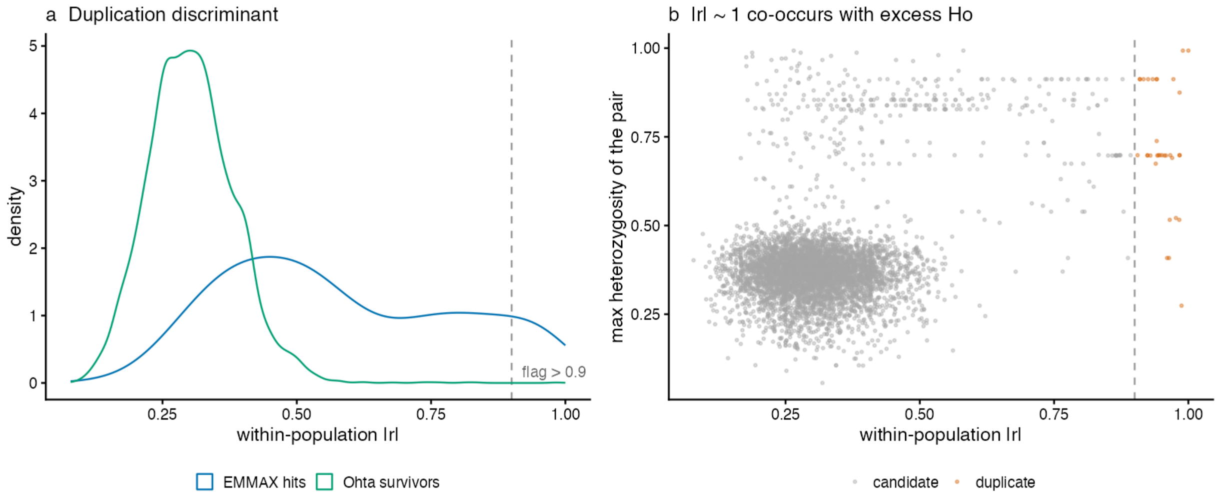

Figure 2: The paralogy discriminant. **(a)** Distribution of within-population $|r|$ for candidate pairs: the among-population survivors (green) concentrate at moderate $|r|$ with a thin tail, whereas the EMMAX hits (blue) carry a heavy tail to $|r| \approx 1$ — the duplicates. Dashed line: the 0.9 flag. **(b)** Pairs at $|r| \approx 1$ (orange, flagged) also carry elevated heterozygosity, corroborating a duplication origin; excess H_o alone is not sufficient (many high-H_o pairs at moderate $|r|$ are genuine).

The cis/trans logic. Consider two loci A and B, each carrying *F. aquilonia* or *F. polyctena* ancestry. There are two parental combinations, ($F.\ aquilonia_A$,$F.\ aquilonia_B$) and ($F.\ polyctena_A$,$F.\ polyctena_B$), and two recombinant ones, ($F.\ aquilonia_A$,$F.\ polyctena_B$) and ($F.\ polyctena_A$,$F.\ aquilonia_B$). A Dobzhansky–Muller incompatibility makes some of these unfit, and each replicate, as it sorts, fixes a *compatible* pair. Two architectures follow:

- If the two *parental* combinations are the compatible ones (the classic case: derived alleles from the two lineages are mutually incompatible, so the recombinants are unfit), each population fixes both-*F. aquilonia* or both-*F. polyctena*. Across replicates the two loci fix the *same* ancestry direction, so their per-population frequencies are positively correlated: **cis** coupling, $R_{\mathrm{st}} > 0$.

- If instead the *recombinant* combinations are the compatible ones (co-adapted heterospecific alleles, so each parental pairing is unfit), populations fix *F. aquilonia* at one locus with *F. polyctena* at the other. The per-population frequencies are *negatively* correlated: **trans** coupling, $R_{\mathrm{st}} < 0$.

Why the sign matters for the confound. A shared ancestry gradient makes *every* locus track overall ancestry, which produces *positive* correlations between loci. So among general loci, cis ($R_{\mathrm{st}} > 0$) co-sorting is indistinguishable from the gradient and cannot on its own indicate an incompatibility. *Trans* co-sorting ($R_{\mathrm{st}} < 0$) is the opposite of what the gradient produces — it requires two loci to fix *opposite* ancestries in the same populations — so it cannot arise from the shared gradient at all. Negative among-population covariance between unlinked loci is therefore the most confound-resistant DMI signature we have. Restricting to bidirectional loci additionally rescues the cis direction: because these loci do not track the gradient, even their positive co-sorting is pair-specific rather than a by-product of overall ancestry.

Test and null. For each bidirectional representative we compute its *F. aquilonia*-oriented allele frequency in each of the twenty populations, and for every unlinked pair (different chromosome, or same chromosome > 10 cM) the correlation R_{st} of those twenty-value vectors. Because bidirectional loci are approximately orthogonal to the ancestry axis, a per-locus *permutation* null — independently shuffling each locus's twenty per-population values, which preserves each locus's marginal sorting but destroys any co-sorting between loci — is a reasonable pre-simulation calibration here (it would not be, for the ancestry-tracking loci of the general screen). We compare the observed number of strongly co-sorting unlinked pairs to that null. Cross-chromosome duplicates are filtered as in Section 3 (only 3 of 178,067 pairs — bidirectional loci are essentially never paralogs).

Result: a real but modest, diffuse excess. There are 276 unlinked pairs at $|R_{\mathrm{st}}| \geq 0.7$, against a permutation-null mean of 165 (95% $\leq$ 186), empirical $p < 0.001$ (Fig. 3b) — about 2.3× the general differentiated-pair rate. Three features make this the most incompatibility-consistent result in Module D so far: it is orthogonal to the ancestry axis; 117 of the 276 are *trans* (negative R_{st}, which the gradient cannot produce; 159 cis); and it is genome-wide, not driven by the Chr24 block (only 6/276 pairs involve Chr24, and the non-Chr24 rate is identical). The honest caveats are equally clear: the *distribution* of R_{st} barely shifts from the null (Fig. 3a) — the excess is a tail-count effect of order a hundred pairs; the candidate network is diffuse (a hairball, not a few sharp modules, Fig. 3c); and the permutation null does not model the shared drift among populations, so it *overstates* significance. The true excess-over-neutral is smaller and awaits the simulation. This is a lead consistent with a weak, distributed contribution of equal-fitness incompatibilities, not a resolved set of strong pairwise DMIs.

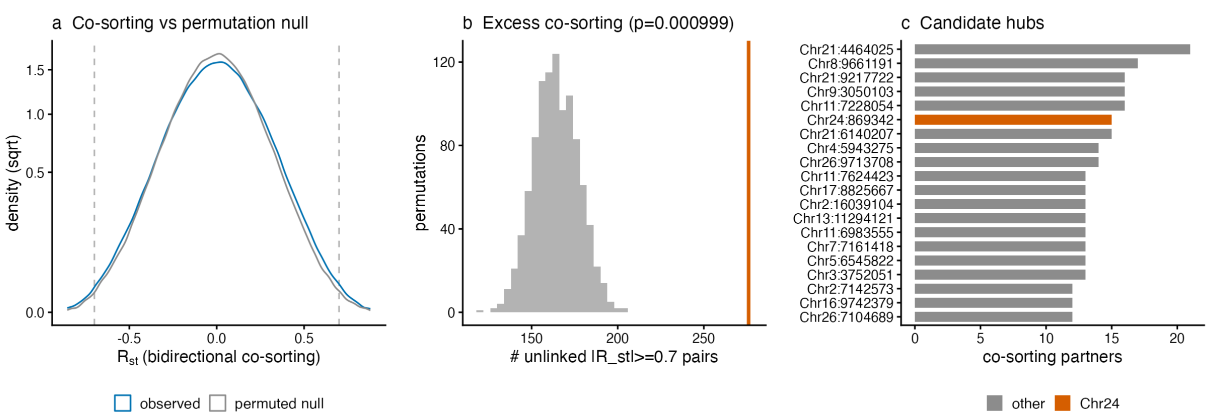

Figure 3: Bidirectional co-sorting test. **(a)** The observed distribution of per-pair R_{st} (blue) is only slightly heavier-tailed than the permutation null (grey). **(b)** But the count of unlinked pairs at $|R_{\mathrm{st}}| \geq 0.7$ (orange line) sits far beyond the permutation-null distribution of that count, $p < 0.001$. **(c)** The candidate network's hub loci are spread across chromosomes; the Chr24 block (orange) is not special.

5 Cluster independence: a by-product of the Manhattans

The mixed-model Manhattans double as a direct test of whether the complexity reduction produced *independent* units. If it did, a focal cluster should give a single local peak — its self-association —

with flat neighbours; several significant peaks in the same region, as individual SNPs would give, would mean the local LD block had been split into clusters that remain correlated. Because the random effect is fitted leave-one-chromosome-out, the focal chromosome is not in its own K, so this local structure is seen honestly. We read it off the focal chromosome with the self-locus omitted (its self-$r^2 \approx 1$ would compress the scale) and its position marked, plotting the relatedness-corrected r^2 of every other cluster (Fig. 4).

About half of the focal loci are locally clean — a single self-position with no significant neighbour (Fig. 4b). The remainder carry one to eight same-chromosome clusters that clear genome-wide significance within $\sim$1 cM of the focal (Fig. 4a,c), but at *moderate* relatedness-corrected r^2 — at most ≈ 0.3, never near one. The clustering has therefore broken each local block apart substantially (these neighbours are far from duplicates of the focal) without reducing it to a single unit, which is the expected outcome of a deliberate trade-off. Merging is bounded by two floors: a fidelity floor, beyond which the consensus stops faithfully summarising its members and the eMLG loses its meaning, and a pairwise-correlation floor below which pieces are already near-independent. Where merging stops — most often against the fidelity floor, inside a large coherent block — the residual correlation between the neighbouring clusters lands at $r^2 \lesssim 0.3$. That ceiling is at least as strict as the fixed $r^2 = 0.2$ that conventional LD pruning imposes, and it is measured on a fairer scale: a relatedness-corrected correlation between consensus units that each summarise many markers, rather than a raw correlation between single SNPs. The residual is also purely local, so the one thing it could still distort — counting a single co-sorting event several times because each side is represented by a few correlated clusters — is removed for the long-range screens by the 10 cM distance threshold, which is itself supported by the observed decay of among-population LD to the inter-chromosomal baseline. Bounded local dependence and long-range de-duplication are thus handled by two separate devices rather than either alone.

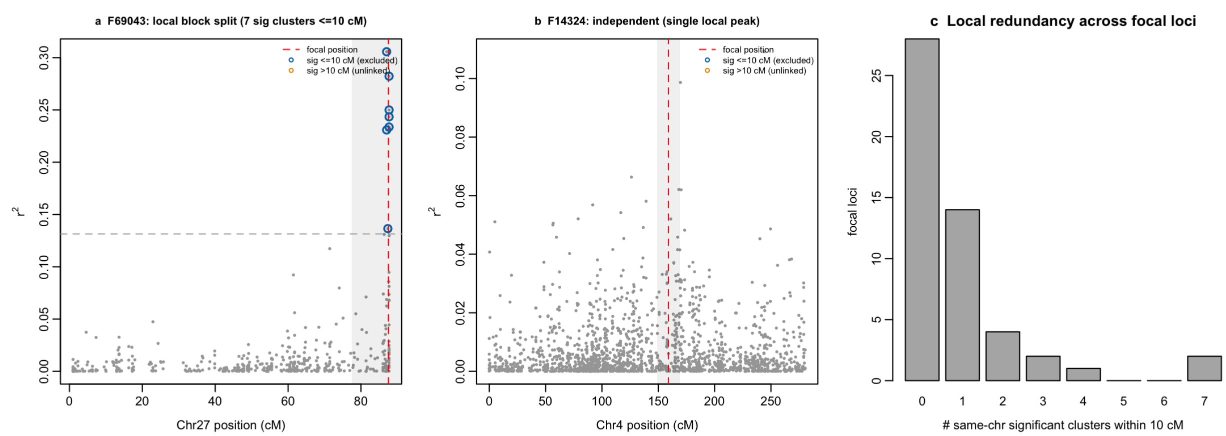

Figure 4: LD-cluster independence, read from the focal chromosome (self-locus omitted; dashed red line = its position; grey band = ±10 cM; dashed grey line = the Bonferroni threshold expressed in r^2; blue = significant within 10 cM, orange = beyond). **(a)** A locus whose local block is split into several correlated clusters, all at $r^2 \lesssim 0.35$ and all within 10 cM. **(b)** A locally independent locus: a single self-position, no significant neighbour. **(c)** Across the focal loci, most have no or few significant same-chromosome neighbours within 10 cM.

6 Following the trans signal: hubs, structure, and the pair-specific residue

The confound-free part of the signal is the *trans* associations (Section 4), so we followed the strongest of them to their source. Most do not come from isolated pairs. They come from *hubs*: a single focal cluster that anti-sorts with many unlinked partners which, in turn, co-vary strongly with one another. The clearest is F33028 (Chr10), with 34 trans partners that correlate among themselves at mean $|r| = 0.66$ — one coherent axis with F33028 at one pole and the partner module at the other, not thirty-four independent incompatibilities. Two properties place it: it is robust to leaving out any single population (so it is not a Karsikas / single-colony artifact), and it survives conditioning on the top genome PCs (so it is not one of the dominant, climate-aligned structure axes — its loading on those is only $r \approx 0.27$). It is instead a *specific, lower-variance* co-sorting module with a broad footprint (a ∼5.5 Mb Chr26 block in trans-LD with a Chr10 anchor; Fig. 5). Its source is the recombination landscape. F33028 sits at 0.4 cM/Mb — the first recombination percentile against a genome-wide median of 20.5 — and 33 of its 37 module clusters fall in the low-recombination tail, the centromeric and pericentromeric regions the map makes plain. That argues *against* a segregating inversion, whose recombination suppression would be its own rather than the reference centromere's, and for a simpler mechanism: low-recombination regions retain the founding admixture co-ancestry that recombination has long since erased elsewhere, so centromeric blocks on different chromosomes stay correlated in ancestry — in whichever direction they were founded, hence the trans sign. The hub is the genome's low-recombination fraction co-retaining ancestral structure, neither a point incompatibility nor a rearrangement. (The genuine pair-specific candidates below are, by contrast, in normal-to-high recombination, consistent with real signal rather than a low-recombination artifact.)

The genuine pair-specific candidates are therefore the trans edges whose *both* endpoints have low connectivity. Of 78 clean trans pairs, 73 run through a hub. Scoring the remaining five on within-hybrid minor allele frequency and leverage (Table 1) removes the rest of the illusion: the strongest of them are *near-fixed* loci, where the minor allele frequency is too low for real disequilibrium (the MAF–LD bound) and a handful of individuals manufacture the correlation — one pair's $r = -0.44$ collapses to -0.03 when five individuals are dropped. Only two survive as robust, appreciable-frequency, leverage-stable pair-specific edges, and both are weak ($r \approx -0.25$). After every structure control the two-locus screen can apply, that is the entire pair-specific residue.

Table 1: Putative pair-specific (non-hub) trans-DMI pairs from the structure-conditioned scan, with per-cluster diagnostic index (DI), parental allele frequencies (the aqu-oriented allele's frequency in each parent species), within-hybrid minor allele frequency (MAF), the consensus correlation r, and its value after dropping the five most influential individuals. Values are given as (first cluster / second cluster). "candidate" = both loci appreciably polymorphic in the hybrids and r leverage-stable; "near-fixed" = MAF too low for real LD; "leverage" = correlation not stable to individual removal.

Pair	Chr	n_{loci}	DI	p_{aqu}	p_{pol}	MAF	r (r_{-5})	disposition
F12448–F30042	3/9	6/5	$-62/-26$	0.59/0.85	0.42/0.17	0.28/0.32	-0.26 (-0.27)	candidate
F67262–F38669	26/12	7/8	$-29/-42$	1.00/0.81	0.65/0.46	0.40/0.17	-0.23 (-0.24)	candidate
F22685–F32802	6/9	5/9	$-42/-15$	0.99/0.88	0.55/0.34	0.07/0.09	-0.70 (-0.55)	near-fixed
F26284–F58007	7/20	5/6	$-52/-42$	0.30/1.00	0.00/0.64	0.05/0.09	-0.44 ($+0.03$)	near-fixed
F37861–F973	11/1	5/32	$-95/-37$	0.50/0.75	0.12/0.47	0.21/0.18	-0.27 (-0.18)	leverage

7 Synthesis and what the simulation will settle

Three structure-corrected views now agree, and they agree on a largely negative result. The among-population screen showed the raw signal is the ancestry gradient. The bidirectional test, working orthogonally to that gradient, found only a modest and diffuse excess of co-sorting. And the mixed model, once its signal is followed to the end — paralogy filtered, dominant structure conditioned out, hubs recognised as low-recombination regions co-retaining ancestral structure rather than incompatibilities, and the survivors screened for the frequency and leverage artifacts the MAF–LD bound invites — leaves only two weak pair-specific trans edges. Each layer of control removed an apparent signal and exposed it as structure, assembly, or a handful of individuals; what remains is marginal.

This is the expected place to be *before* the null. Every control here removes structure that is *aligned* with what it conditions on, and none of them can say whether the weak residue exceeds what neutral, recombination-matched drift would produce in twenty replicate populations of this size and sampling design. That is the one thing Module E is built to answer, and it is now the load-bearing next step: it turns "a weak residual signal" into "an excess over neutral", or not. The handful of genuine pair-specific candidates in Table 1 are then few and weak enough to be examined one at a time once the null gives them a scale.

One loose end is resolved rather than deferred. It was tempting to assign the hub axes (F33028 and kin) to the *extrinsic* climate side, on the strength of an individual-level correlation between the dominant genome axes and the climate PCs. The climate-association test says otherwise. Against the BayPass genotype–environment scan, F33028 and its module are among the *least* climate-associated loci in the genome — the module's median Bayes factor is *below* the genome-wide background for both climate axes ($p \sim 10^{-10}$), and F33028 itself falls in the first percentile for the first axis. At the locus level the genome ancestry axes barely track the climate association at all ($r \approx 0.1$), so the individual-level genome–climate correlation is population-level entanglement (ancestry and climate both vary with geography), not locus-specific climate selection. The hubs therefore belong to *neither* arm — not a resolved intrinsic incompatibility, and not an extrinsic climate response, but the genome's low-recombination (centromeric) fraction retaining founding admixture co-ancestry (Section 6): a recombination-landscape feature that a two-locus ancestry-LD screen surfaces and that neither the sorting nor the climate analysis is built to catch.

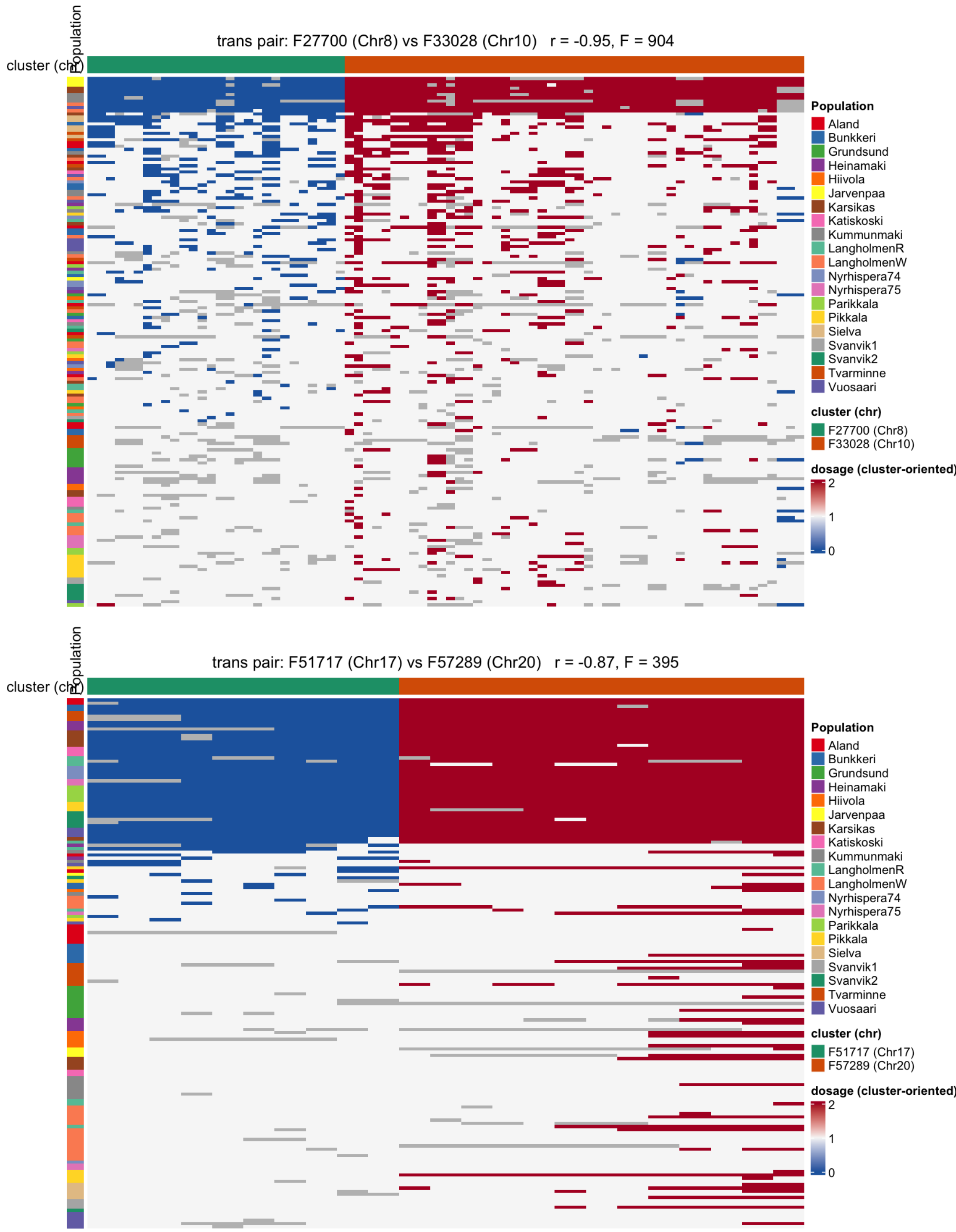

Figure 5: Two of the strongest trans pairs, as genotype heatmaps of the two clusters' member markers (individuals in rows, ordered by the first cluster's ancestry; each cluster's markers polarised to its own consensus so a trans pair renders as a mirror image, not a flipped-together block). In the sorted individuals (the coloured band) the two unlinked clusters fix *opposite* ancestries — blue at one, red at the other — while the heterozygous majority sits pale below. These are strong and clean, but both belong to the low-recombination hub structure (top: F33028), not to the pair-specific residue: they show what co-retained ancestral co-ancestry looks like, not an incompatibility.